

Day 15 (24/11/2023)

Q) Print prime numbers from given range (1,10) with loops.

```
for i in range(2,11,1):  
    for j in range(2,i,1):  
        if (i%j==0):  
            break  
    else:  
        print(i)
```

2
3
5
7

```
for i in range(1,11,1):  
    if (i!=1):  
        for j in range(2,i,1):  
            if (i%j==0):  
                break  
        else:  
            print(i)
```

2
3
5
7

Q) Print prime numbers from given range (1,10) with function.

```
def prime_series(x1,x2):  
    for i in range(x1,x2+1,1):  
        if (i!=1):  
            for j in range(2,i,1):  
                if (i%j==0):  
                    break  
            else:  
                return i
```

prime_series(1,10)

2 ⇒ only you get 2 because
it is breaking there & returning
only 2.

```
def prime_series(x1,x2):  
    l1 = []  
    for i in range(x1,x2+1,1):  
        if (i!=1):  
            for j in range(2,i,1):  
                if (i%j==0):  
                    break  
            else:  
                l1.append(i)  
    return l1
```

prime_series(1,10)

o/p [2, 3, 5, 7]

Function as Parameter - $\#$ we are giving function as parameter.

In python functions are first class objects.

it means

we can assign them to variables.

we can pass them as parameters to other functions.

we can return them from functions.

ex-

```
def square(x):  
    return x*x
```

```
def cube(x):  
    return x*x*x
```

```
def apply_fun(fun, num):  
    return fun(num)
```

calling fun
apply_fun(square, 5)

o/p 25

Nested functions - A nested function is a function defined inside another function.

it helps in encapsulation and modularity.

syntax

```
def outer_fun(name):  
    def inner_fun():  
        return f"hello, {name}"  
    return inner_fun()
```

outer_fun("poojyy")

'hello, poojyy'

key points-

the inner function can access variables from the outer functions (called closure)

nested functions are often used in decorators and encapsulation.

recursion functions- It is a function that calls itself, either directly or indirectly, to solve a problem by breaking it down into smaller, simpler subproblems.